

FUNDAMENTALS · WHY BEFORE WHAT · COMPREHENSIVE LECTURE NOTES

Module 1: How Internet Works

The internet was not invented. It was argued into existence, one problem at a time.

Author & Educator Notes · Submodule 01 · Target Audience: CS / Web Engineering Students

Core Pedagogical Rule: Every concept in this topic — IP addresses, packet switching, domain names, DNS, routing, ISPs — exists because something broke in the real world. **Teach the break first.** When students feel the pain of the broken state, the technical solution feels natural and inevitable instead of arbitrary vocabulary to memorize.

UCLA → STANFORD RESEARCH INSTITUTE · 29 OCT 1969 · 22:30 PST

L O _

[!] System crashed before typing the "G". The first message in internet history was "LO".

1 1943–1946: One Computer, and No Reason to Connect It

Begin where there is zero networking problem. This stage exists purely to establish the baseline and create stark contrast with what happens next.

ENIAC — THE FIRST GENERAL-PURPOSE ELECTRONIC COMPUTER

WHERE

Moore School of Electrical Engineering, **University of Pennsylvania**, Philadelphia, USA.

WHEN

Constructed 1943–1945; publicly unveiled on **14 February 1946**.

WHO

John Mauchly (physics professor) and **J. Presper Eckert** (graduate student).
Not a giant corporate lab — two individuals and a university contract.

WHY

Commissioned by the **U.S. Army** during WWII to calculate artillery shell firing trajectories. Its first classified mission was thermonuclear calculations finished in 2 hours (work that took humans years).

SCALE

30 tons, 1,500 sq ft room, 40 cabinets (9 ft high), containing over **18,000 vacuum tubes**.

PROGRAMMING

No keyboard, no monitor, no OS. Reprogrammed by physically re-plugging patch cables and flipping hundreds of switches — a multi-day task executed by a team of pioneering women mathematicians.

THE LESSON OF THIS BEAT

You have enormous computing horsepower, and **zero connectivity**. Not because networking

was technically hard — but because there was literally no other computer in existence to connect it to! Let the students appreciate this stillness before you break it.

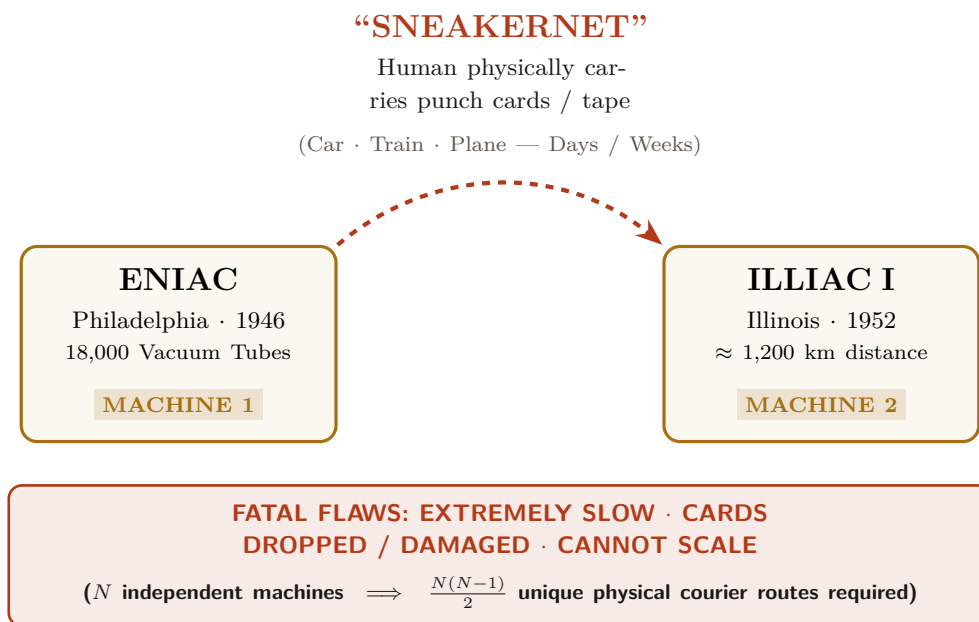
▪ **BOARD / CAMERA CUE**

Draw exactly one large box on the center of the board. Label it: [ENIAC · Philadelphia · 1946]. Leave the entire rest of the board completely empty. **That emptiness is the lesson.**

2 1950s: The Second Computer Is Built — Sneakernet Arrives

*Through the late 1940s and 1950s, universities built independent machines. Example: **ILLIAC I** at the University of Illinois (1 Sept 1952) — the first computer owned entirely by an educational institution.*

Now there are two machines, roughly **1,200 km apart**. Both produce calculations the other needs. And there is only one way to move data: **a human physically carries punch cards or magnetic tape in a bag via train, plane, or car.**



■ ANALOGY FOR THE LECTURE

“Socho, aaj agar aapko dost ko ek file bhejni ho, aur aapko Bhopal se Bangalore train pakadni pade har baar! Ye 1950s ki reality thi. Aur yahi wo physical takleef thi jisne computer networking ko janm diya.”

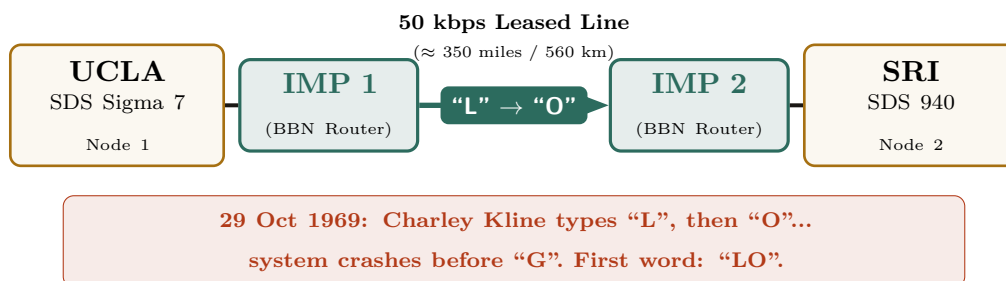
Transition Question to Class: *“Toh sabse pehla natural solution kya dimag mein aayega? In dono machines ko ek taar (wire) se jod diya jaye!”*

3 1966–1969: ARPANET — Wiring Computers Together

*The fix was funded by the US Department of Defense, but **not** primarily for nuclear war survival (a popular myth). It was built because multi-million-dollar supercomputers sat idle in universities while other researchers waited months for compute access.*

KEY ARCHITECTS AND ARTIFACTS OF ARPANET

FUNDER	ARPA (Advanced Research Projects Agency, DoD) — created in 1958 in response to USSR's Sputnik.
THEORY	Leonard Kleinrock (UCLA) created queuing theory and packet networking mathematical models. Paul Baran (RAND) conceived robust distributed packet switching.
PLAN	Larry Roberts published the formal ARPANET Program Plan in June 1968.
HARDWARE	BBN (Bolt, Beranek and Newman) won the \$1M contract to build the IMP (Interface Message Processor) — the ancestor of modern routers.
FIRST 4 NODES	1. UCLA 2. Stanford Research Institute (SRI) 3. UC Santa Barbara 4. Univ. of Utah.
PIONEERS	Steve Crocker authored RFC 1. Jon Postel managed internet numbering/DNS assignments for decades.



WHY THE IMP IS HISTORICALLY CRUCIAL

Host mainframes were expensive and refused to waste CPU cycles managing network handshakes. BBN built dedicated intermediate hardware (IMPs) whose sole job was routing. **The modern router exists because host computers refused to do the networking job.**

4 The Core Paradigm: Circuit Switching vs Packet Switching

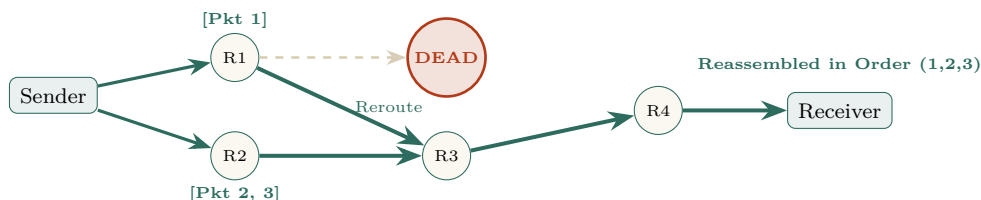
This is the conceptual backbone of all computer networks. Always teach it as a direct comparison against the telephone network of the 1960s.

1. CIRCUIT SWITCHING (THE TELEPHONE WAY — DEDICATED & FRAGILE)



× Single reserved physical copper circuit. If intermediate node fails, **entire call terminates immediately**.

2. PACKET SWITCHING (THE INTERNET WAY — CHOPPED, ADAPTIVE & RESILIENT)



Attribute	Circuit Switching (Landline Phone)	Packet Switching (Internet)
Path Reservation	Dedicated fixed route held open for full call	No reservation; dynamic per-packet routing
Efficiency	Line wasted during silence/pauses	Multiple users interleave packets on shared lines
Failure Recovery	Fatal: if one switch fails, conversation ends	Resilient: packets simply route around dead nodes
Order Guarantee	In-order by physical nature	Out-of-order arrival handled via sequence numbers

■ CLASSROOM ANALOGY

“Ek 200 page ki kitaab Bangalore bhejni hai:

Option A (Circuit): Poori kitaab ek truck mein daal di. Agar raaste mein truck ka accident hua, toh kitaab hamesha ke liye kho gayi.

Option B (Packet): Har page pe serial number likha, 200 alag lifafon mein daala, alag-alag postmen aur raaston se bheja. Kuch late pahunchenge, kuch jaldi — par receiver number dekh kar wapas arrange kar lega. Missing page ko dobara maang lenge. **Internet always chooses Option B.**”

5 1973–1974: TCP/IP & The Birth of “The Internet”

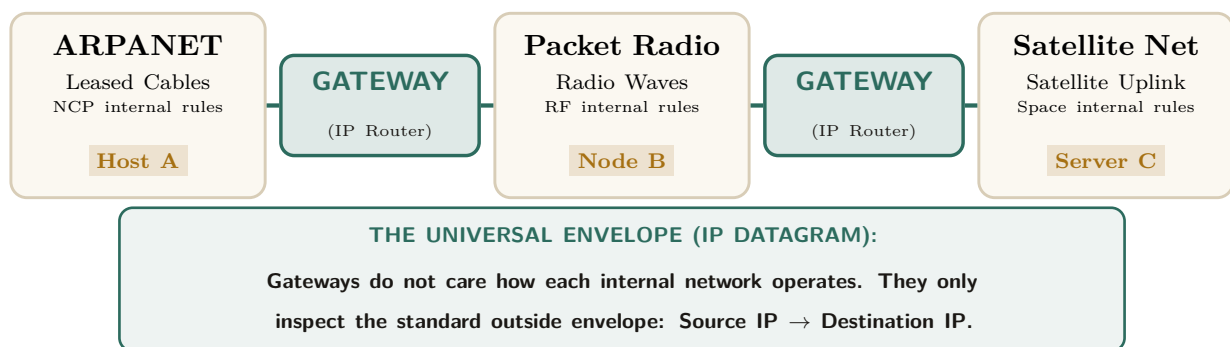
THE NEW CRISIS: ISLANDS OF INCOMPATIBLE NETWORKS

By 1973, ARPANET was not alone. Hawaii had **ALOHANET** (radio packets), Europe had **CYCLADES**, and satellite networks were launching.

The Wall: Inside ARPANET, machines talked using NCP. Inside Packet Radio, machines spoke their own radio dialect. **None could communicate across network boundaries.** We were back to the 1952 problem — except the islands were now entire networks!

TCP/IP — THE UNIVERSAL INTER-NETWORKING PROTOCOL

INVENTORS	Vinton Cerf (UCLA/Stanford) and Robert Kahn (ARPA/BBN).
ORIGINS	Designed in a Palo Alto hotel room in 1973 ; published May 1974: “ <i>A Protocol for Packet Network Intercommunication</i> ”.
ETYMOLOGY	Cerf and Kahn coined the word “ Internet ” — an abbreviation of inter-networking of networks .
ARCHITECTURE	Introduced Gateways (routers) at borders and Datagrams (envelopes with a standard IP header). Internal networks could run whatever hardware they wanted.
FLAG DAY	1 January 1983: ARPANET permanently turned off NCP and made TCP/IP mandatory. This is widely considered the technical birthday of the Internet.



■ CAMERA PUNCHLINE

“Internet ka matlab hi yehi hai — *Inter-networking of networks*. Alag-alag networks ko aapas mein jodna bina unke internal rules ko badle. Isiliye border pe translator (Gateway) baithaya aur ek universal postal address banaya: **The IP Address.**”

6 1983: Domain Names & DNS — Breaking the Phonebook

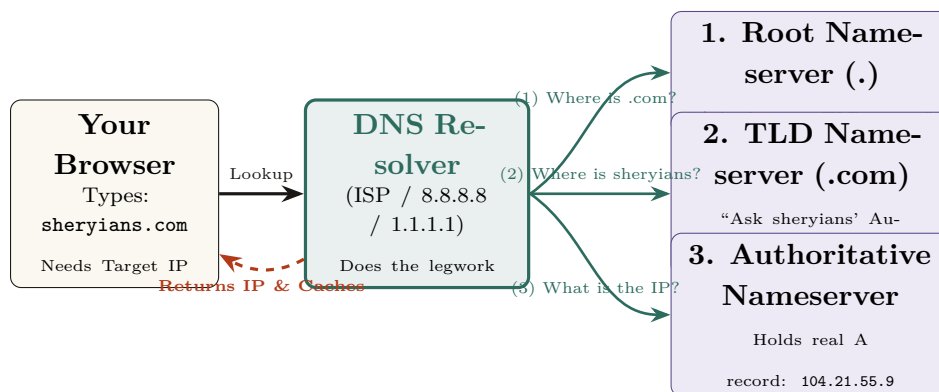
THE PRE-DNS REALITY: HOSTS.TXT AT SRI

Before DNS, name resolution was handled by a single file called `HOSTS.TXT` hosted on one computer at SRI in California.

- Every night, all computers downloaded the latest copy via FTP.
- To register a new hostname, you literally **called SRI on the telephone during California business hours**.
- **The Crash Point:** As thousands of hosts joined, the file exploded in size, local copies went stale, name collisions erupted, and SRI became a single global bottleneck.

DNS — THE HIERARCHICAL DISTRIBUTED DATABASE (RFC 882 / 883)

INVENTOR	Paul Mockapetris at USC Information Sciences Institute (assigned by Jon Postel).
YEAR	1983 (RFC 882/883). Still running the modern world with near-zero fundamental architecture changes.
CORE IDEA	Replaced one flat list with a tree hierarchy . Authority is delegated downwards: the Root server delegates <code>.com</code> to a registry, which delegates <code>sheryians.com</code> to its owner.



■ THE PHONEBOOK VS TREE ANALOGY

“Log DNS ko ‘Internet ki phonebook’ bolte hain. Yeh bolna galat nahi hai, par adhoora hai. Phonebook ek single kitaab hoti hai ek table par. DNS ek ped (tree) hai, jiski har branch alag insaan sambhalta hai. Isiliye yeh 1983 se aaj tak bina ruke scale kar raha hai.”

7 1986–1995: ISPs, Commercialization & Hop-by-Hop Routing

*Until 1991, commercial businesses were strictly **illegal** on the internet backbone. The transition to our modern internet was driven by policy changes as much as engineering.*

THE COMMERCIALIZATION ARC

1986	NSFNET launched as the high-speed academic backbone. Connected 2,000 computers in 1986 → over 2 million by 1993.
THE RULE	NSFNET enforced an Acceptable Use Policy (AUP) banning all commercial/for-profit activity.
1991	NSF lifted the commercial ban. The ISP industry was born (companies selling access directly to consumers and businesses).
1995	NSF decommissioned its public backbone. The internet became a global mesh of private networks interconnected by Peering (free exchange) and Transit (paid routing).



ROUTING IS A SERIES OF LOCAL DECISIONS, NOT A FIXED MAP:

No single entity controls or owns this entire path.
90,000+ autonomous networks negotiate hops in real time.

▪ LIVE DEMONSTRATION IN CLASS

Open your command prompt live on screen and run:

```
tracert google.com (Windows) or traceroute google.com (Mac/Linux)
```

Show students every hop IP address printing line by line. It turns abstract theory into tangible reality.

8 1990+: The Web on Top of the Plumbing (Web 1.0 → 3.0)

***Crucial Distinction:** The Internet is **not** the Web. The Internet is the underlying road and packet delivery infrastructure (1969–1990). The Web is the first massive vehicle built to drive on that road (1990+).*

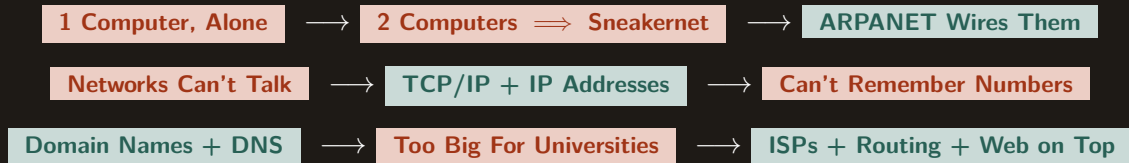
- **Web 1.0 (Read-Only · 1990–2000):** Invented by **Tim Berners-Lee** at CERN (HTML, HTTP, URLs). Static pages hosted by universities and hobbyists. Users were purely passive readers.
- **Web 2.0 (Read-Write / Interactive · 2000–Present):** Databases, dynamic AJAX, user authentication, social media, uploads. Users became creators. (Everything students build in modern web dev lives here).
- **Web 3.0 (Read-Write-Own · Emerging/Contested):** Decentralized protocols, cryptographic identity, blockchains. (Be honest with students: this is an evolving area with active debates, not settled infrastructure like TCP/IP).

9 Recap: Master Timeline & The One-Line Spine

Year	Historical Milestone	The Specific Problem It Solved
1943–46	ENIAC built at UPenn	None — no networking problem existed yet.
1952	ILLIAC I & multi-computer era	Created the problem: data had to be carried by hand (Sneakernet).
1968–69	ARPANET & BBN IMPs	Allowed computers to exchange data directly without human couriers.
1969	Packet Switching deployed	Prevented single-line failure from killing entire communications.
1973–74	Cerf & Kahn design TCP/IP	Allowed completely incompatible networks to interconnect (The Internet).
1983	Paul Mockapetris creates DNS	Replaced fragile HOSTS.TXT with a globally scalable hierarchy.
1986	NSFNET backbone launched	Provided high-speed academic bandwidth across North America.
1989–90	Tim Berners-Lee invents WWW	Created an accessible hypertext application on top of the internet.
1991	NSF lifts commercial restriction	Enabled the creation of consumer ISPs and private web hosting.
1993	Mosaic Browser released	Brought inline images and consumer usability to the web.
1995	NSFNET decommissioned	Internet transitioned entirely to private peering and transit.

The One-Line Spine of the Whole Lecture

CORE TAKEAWAY CHAIN FOR THE BOARD



■ ONE HONEST LOCAL CONTEXT TO MENTION (INDIA)

While this history is US-centered, public internet in India arrived on **15 August 1995**, when **VSNL** launched public dial-up service. Mention this: *“Aapke parents ke college ke time India mein internet nahi tha — it is this recent!”*

■ CLOSING BRIDGE TO NEXT EPISODE (TRANSPORT LAYER: TCP VS UDP)

*“Lekin dhyan do — abhi tak humne sirf IP address ke zariye data packets ko router-to-router pahunchana seekha hai. Lekin agar packet raaste mein drop ho jaye, ya order ulta-pulta ho jaye, toh kya hoga? Aur realtime audio/video ke liye hume fast stream chahiye ya 100% reliable stream? Yeh hai hamara agla topic: **Transport Layer: TCP vs UDP.**”*